SQL JOINS - HANDS-ON PRACTICE

Scenario-based exercises with queries and expected results

Use these exercises after learning the basic JOIN concepts. Each scenario gives sample data, a real requirement, the SQL query, and the expected result.

Common Practice Data

EMPLOYEES

emp_id	emp_name	dept_id	salary
1	Arun	101	50000
2	Priya	102	70000
3	Rahul	101	60000
4	Sneha	103	80000
5	Karthik	NULL	45000

DEPARTMENTS

dept_id	dept_name
101	IT
102	Finance
103	HR
104	Marketing

1. INNER JOIN - Employees with a valid department

Requirement: Show only employees whose department exists in the DEPARTMENTS table.

Query:

```
SELECT e.emp_id, e.emp_name, d.dept_name
FROM employees e
INNER JOIN departments d
    ON e.dept_id = d.dept_id;
```

Expected result / idea:

```
emp_id | emp_name | dept_name
1 | Arun  | IT
2 | Priya | Finance
3 | Rahul | IT
4 | Sneha | HR
```

Why: Karthik is not returned because his dept_id is NULL.

2. LEFT JOIN - Show every employee

Requirement: Display all employees with their department name. Employees without a department must also appear.

Query:

```
SELECT e.emp_id, e.emp_name, d.dept_name
FROM employees e
```

```
LEFT JOIN departments d
    ON e.dept_id = d.dept_id;
```

Expected result / idea:

```
emp_id | emp_name | dept_name
1 | Arun    | IT
2 | Priya   | Finance
3 | Rahul   | IT
4 | Sneha   | HR
5 | Karthik | NULL
```

Why: LEFT JOIN keeps every row from EMPLOYEES.

3. RIGHT JOIN - Show every department

Requirement: Display every department and any employee working in it. Departments with no employees must still appear.

Query:

```
SELECT e.emp_name, d.dept_name
FROM employees e
RIGHT JOIN departments d
    ON e.dept_id = d.dept_id;
```

Expected result / idea:

```
emp_name | dept_name
Arun  | IT
Rahul | IT
Priya | Finance
Sneha | HR
NULL  | Marketing
```

Why: Marketing appears even though no employee belongs to it.

4. FULL OUTER JOIN - Keep unmatched rows from both sides

Requirement: Show all employees and all departments, including employees without departments and departments without employees.

Query:

```
SELECT e.emp_name, d.dept_name
FROM employees e
FULL OUTER JOIN departments d
    ON e.dept_id = d.dept_id;
```

Expected result / idea:

```
emp_name | dept_name
Arun    | IT
Rahul   | IT
Priya   | Finance
Sneha   | HR
```

```
Karthik | NULL
NULL    | Marketing
```

Why: FULL OUTER JOIN keeps unmatched records from both tables.

5. CROSS JOIN - Every employee and every shift

Requirement: A manager wants every possible employee/shift combination.

Query:

```
SELECT e.emp_name, s.shift_name
FROM employees e
CROSS JOIN shifts s;
```

Expected result / idea:

```
If there are 5 employees and 3 shifts:
5 x 3 = 15 result rows
```

Why: CROSS JOIN creates every possible combination.

6. SELF JOIN - Employee and manager

Requirement: An EMPLOYEES table also has manager_id. Show each employee with the manager's name.

Query:

```
SELECT e.emp_name AS employee,
       m.emp_name AS manager
FROM employees e
LEFT JOIN employees m
    ON e.manager_id = m.emp_id;
```

Expected result / idea:

```
employee | manager
Priya    | Arun
Rahul    | Arun
Sneha    | Priya
```

Why: The same table is used twice: e represents employees and m represents managers.

7. JOIN + WHERE - High salary employees in IT

Requirement: Find employees from the IT department whose salary is greater than 55,000.

Query:

```
SELECT e.emp_name, e.salary, d.dept_name
FROM employees e
INNER JOIN departments d
    ON e.dept_id = d.dept_id
WHERE d.dept_name = 'IT'
  AND e.salary > 55000;
```

Expected result / idea:

```
emp_name | salary | dept_name
Rahul | 60000 | IT
```

Why: JOIN connects the tables; WHERE filters individual rows.

8. JOIN + GROUP BY - Employee count by department

Requirement: Show each department and the number of employees in that department, including departments with zero employees.

Query:

```
SELECT d.dept_name,
       COUNT(e.emp_id) AS employee_count
FROM departments d
LEFT JOIN employees e
    ON d.dept_id = e.dept_id
GROUP BY d.dept_name;
```

Expected result / idea:

```
dept_name | employee_count
IT        | 2
Finance   | 1
HR        | 1
Marketing | 0
```

Why: COUNT(e.emp_id) gives 0 for Marketing because no employee matched.

9. JOIN + GROUP BY + HAVING

Requirement: Show only departments having more than one employee.

Query:

```
SELECT d.dept_name,
       COUNT(e.emp_id) AS employee_count
FROM departments d
INNER JOIN employees e
    ON d.dept_id = e.dept_id
GROUP BY d.dept_name
HAVING COUNT(e.emp_id) > 1;
```

Expected result / idea:

```
dept_name | employee_count
IT | 2
```

Why: HAVING filters the grouped result.

10. LEFT JOIN to find missing matches

Requirement: Find employees who are not assigned to any valid department.

Query:

```
SELECT e.emp_id, e.emp_name
FROM employees e
LEFT JOIN departments d
    ON e.dept_id = d.dept_id
WHERE d.dept_id IS NULL;
```

Expected result / idea:

```
emp_id | emp_name
5 | Karthik
```

Why: This LEFT JOIN + IS NULL pattern is commonly used to find unmatched records.

11. Three-table JOIN - Employee, department and project

Requirement: Assume PROJECTS has project_id, project_name and dept_id. Show employee, department and project.

Query:

```
SELECT e.emp_name,
       d.dept_name,
       p.project_name
FROM employees e
INNER JOIN departments d
    ON e.dept_id = d.dept_id
INNER JOIN projects p
    ON d.dept_id = p.dept_id;
```

Expected result / idea:

```
Practice point: trace one relationship at a time:
EMPLOYEES -> DEPARTMENTS -> PROJECTS
```

Why: This is useful practice for real project queries involving more than two tables.

12. Interview Challenge - Department with average salary

Requirement: Show department name and average employee salary. Return only departments where average salary is above 55,000.

Query:

```
SELECT d.dept_name,
       AVG(e.salary) AS avg_salary
FROM departments d
INNER JOIN employees e
    ON d.dept_id = e.dept_id
GROUP BY d.dept_name
HAVING AVG(e.salary) > 55000;
```

Expected result / idea:

```
dept_name | avg_salary
Finance | 70000
HR      | 80000
```

Why: IT average is (50000 + 60000) / 2 = 55000, so it does not satisfy > 55000.

Practice Without Looking at the Answer

1. Find all employees who do not have a department.

2. Find all departments that currently have no employees.

3. Show employee name, department name and salary only for Finance and IT.

4. Show total salary paid by each department.

5. Show departments where total salary is greater than 100,000.

6. Show the highest-paid employee salary in each department.

7. Show every employee with every available shift.

8. Using a self join, show employee name and manager name.

9. Join three tables: employees, departments and projects.

10. Find departments having at least two employees earning more than 50,000.

Quick Memory

JOIN	Remember
INNER JOIN	Only matching rows
LEFT JOIN	All left + matching right
RIGHT JOIN	All right + matching left
FULL OUTER JOIN	All rows from both sides
CROSS JOIN	Every possible combination
SELF JOIN	Join a table to itself